

A positive ℓ_p -sum class for fully Mackey complete duals

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Status

This packet records a claimed partial positive answer to Problem 4.15 of Guirao, Martinez-Cervantes and Rodriguez [1]. The claim is not a full solution of the problem; it proves the desired implication for a broad nonseparable class of ℓ_p -sums.

Source problem

Guirao, Martinez-Cervantes and Rodriguez [1, Problem 4.15] ask:

Is X^* fully Mackey complete whenever X is w^* -sequentially dense in X^{**} ?

They prove the affirmative answer when X is separable [1, Corollary 4.8], and when X is Asplund [1, Corollary 4.9]. Their Remark 4.7 notes that a negative answer would have to use a nonseparable, non-Asplund space without subspaces isomorphic to ℓ_1 .

Proof Intuition

The obstruction in the nonseparable case is that a norming closed subspace $Y \subset X^{**}$ need not project cleanly onto the countable coordinates on which a given $u \in X^{**}$ is supported. The main observation is that, for ℓ_p -sums of separable spaces, one can enlarge the countable support of u to another countable set A so that the actual intersection $Y \cap X_A^{**}$ is still norming for X_A^* . This is done recursively. At each countable stage, the dual ball of the countable sub-sum is weak-star compact metrizable, so a countable family of elements of Y suffices to keep the norming inequality on that stage; adding their countable supports prepares the next stage.

Once this localization is available, the problem becomes separable. The countable sub-sum contains no ℓ_1 , so the source paper's separable criterion says its dual is fully Mackey complete. The resulting compact LK witness lives entirely on the countable coordinates and is therefore also a valid witness in the original large sum.

Notation and source facts

For a Banach space Z , write Z as canonically embedded in Z^{**} . For a dual pair $\langle Z^*, Z^{**} \rangle$, weak-star means $\sigma(Z^{**}, Z^*)$.

We use the characterization from [1, Theorem 3.5]: a Banach space W is fully Mackey complete if and only if it has property (LK) : for every norming norm-closed subspace $Y \subset W^*$ and every $u \in W^* \setminus Y$, there is an absolutely convex weak-star compact $K \subset W^*$ such that

$$K \subset Y \oplus [u], \quad u \in \overline{K \cap Y}^{w*}.$$

We also use [1, Corollary 4.8]: if Z is separable, then Z^* is fully Mackey complete if and only if Z contains no copy of ℓ_1 .

Main theorem

Theorem 1. *Let $1 < p < \infty$, let Γ be a set, and let $(E_\gamma)_{\gamma \in \Gamma}$ be separable Banach spaces containing no subspace isomorphic to ℓ_1 . Put*

$$X = \left(\bigoplus_{\gamma \in \Gamma} E_\gamma \right)_{\ell_p}.$$

*Then X is weak-star sequentially dense in X^{**} , and X^* is fully Mackey complete.*

Remark 1. *If Γ is uncountable and at least one E_γ is separable non-Asplund without ℓ_1 , for example a James-tree type space, then this gives a nonseparable and non-Asplund positive class for Problem 4.15.*

Preliminary lemmas

For $A \subset \Gamma$, write

$$X_A = \left(\bigoplus_{\gamma \in A} E_\gamma \right)_{\ell_p}.$$

We identify X_A , X_A^* , and X_A^{**} with the corresponding coordinate subspaces of X , X^* , and X^{**} . Let $P_A : X^{**} \rightarrow X_A^{**}$ be the coordinate projection. Every element of X^{**} has countable support because $1 < p < \infty$.

Lemma 1. *If $A \subset \Gamma$ is countable, then X_A is separable and contains no copy of ℓ_1 .*

Proof. Separability follows from countability of A and separability of each E_γ .

To rule out ℓ_1 , use Rosenthal's ℓ_1 theorem. Let (x_n) be a bounded sequence in X_A . Enumerate $A = \{\gamma_1, \gamma_2, \dots\}$. Since each E_{γ_j} contains no ℓ_1 , Rosenthal's theorem gives, by a diagonal argument, a subsequence (x_{n_k}) such that for each j the coordinate sequence $(P_{\{\gamma_j\}} x_{n_k})_k$ is weakly Cauchy in E_{γ_j} .

For any $x^* = (x_\gamma^*)_{\gamma \in A} \in X_A^*$, approximate x^* in norm by a finitely supported functional. The finite-coordinate part is Cauchy on (x_{n_k}) , and the tail is uniformly small because (x_{n_k}) is bounded. Thus $(x^*(x_{n_k}))_k$ is Cauchy for every $x^* \in X_A^*$, so (x_{n_k}) is weakly Cauchy. Rosenthal's theorem now implies that X_A contains no isomorphic copy of ℓ_1 . $\square$

Lemma 2. *Let $Y \subset X^{**}$ be a norming norm-closed subspace. For every countable $A_0 \subset \Gamma$ there is a countable $A \supset A_0$ such that*

$$Y_A := Y \cap X_A^{**}$$

is norming for X_A^ .*

Proof. Since Y is norming, choose $0 < \delta \leq 1$ such that

$$\sup\{|y(x^*)| : y \in Y, \|y\| \leq 1\} \geq \delta \|x^*\| \quad (x^* \in X^*). \quad (1)$$

We recursively build countable sets

$$A_0 \subset A_1 \subset A_2 \subset \dots$$

and countable sets $C_n \subset Y \cap B_{X^{**}}$. Assume A_n is countable. Then X_{A_n} is separable, so $(B_{X_{A_n}^*}, w^*)$ is compact metrizable. For each rational $r \in (0, 1)$, put

$$O_r = \{x^* \in B_{X_{A_n}^*} : \|x^*\| > r\}.$$

For every $x^* \in O_r$, (1) gives $y \in Y \cap B_{X^{**}}$ with $|P_{A_n}y(x^*)| = |y(x^*)| > \delta r$. Hence the weak-star open sets

$$\{z^* \in B_{X_{A_n}^*} : |P_{A_n}y(z^*)| > \delta r/2\}, \quad y \in Y \cap B_{X^{**}},$$

cover O_r . Since O_r is an open subspace of a compact metrizable space, it is Lindelof. Choose a countable subcover. Taking the union over rational $r \in (0, 1)$ gives a countable set $C_n \subset Y \cap B_{X^{**}}$ such that

$$\sup_{y \in C_n} |P_{A_n}y(x^*)| \geq \frac{\delta}{2} \|x^*\| \quad (x^* \in X_{A_n}^*). \quad (2)$$

Now let A_{n+1} be A_n together with the supports of all elements of C_n . This is countable, because each element of X^{**} has countable support.

Set $A = \bigcup_n A_n$. We claim that $Y_A = Y \cap X_A^{**}$ is norming for X_A^* . The case $x^* = 0$ is trivial. Let $x^* \in X_A^*$ be nonzero and write x_n^* for its restriction to A_n , extended by zero to A . Then $\|x^* - x_n^*\| \rightarrow 0$. Fix a small $\eta > 0$ and choose n so large that

$$\|x^* - x_n^*\| < \eta \|x^*\|, \quad \|x_n^*\| > (1 - \eta) \|x^*\|.$$

By (2), choose $y \in C_n$ which is within $\eta \|x^*\|$ of the supremum at x_n^* , so

$$|P_{A_n}y(x_n^*)| > \frac{\delta}{2} \|x_n^*\| - \eta \|x^*\|.$$

Since $y \in C_n \subset Y \cap X_A^{**}$ and $\|y\| \leq 1$,

$$|y(x^*)| \geq |P_{A_n}y(x_n^*)| - |y(x^* - x_n^*)| > \left(\frac{\delta}{2}(1 - \eta) - 2\eta \right) \|x^*\|.$$

Taking η sufficiently small, we get for instance

$$\sup\{|y(x^*)| : y \in Y_A, \|y\| \leq 1\} \geq \frac{\delta}{4} \|x^*\|.$$

Thus Y_A is norming for X_A^* . □

Proof of Theorem 1

First we show that X is weak-star sequentially dense in X^{**} . Let $u \in X^{**}$ and let $A = \text{supp}(u)$, which is countable. By Lemma 1, X_A is separable and contains no copy of ℓ_1 . The source paper's separable criterion [1, Corollary 4.8] implies that $(B_{X_A^{**}}, w^*)$ is Frechet-Urysohn. Since X_A is weak-star dense in X_A^{**} by Goldstine's theorem, a sequence from X_A converges weak-star to u after scaling to a suitable ball. Therefore X is weak-star sequentially dense in X^{**} .

It remains to prove that X^* is fully Mackey complete. We use property (LK) from [1, Theorem 3.5]. Let $Y \subset X^{**}$ be a norming norm-closed subspace and let $u \in X^{**} \setminus Y$. Let $A_0 = \text{supp}(u)$, and apply Lemma 2 to obtain a countable $A \supset A_0$ such that $Y_A = Y \cap X_A^{**}$ is norming for X_A^* .

By Lemma 1, X_A is separable and contains no ℓ_1 . Hence X_A^* is fully Mackey complete by [1, Corollary 4.8]. Since $u \in X_A^{**} \setminus Y_A$, property (LK) for X_A^* gives an absolutely convex weak-star compact set $K \subset X_A^{**}$ such that

$$K \subset Y_A \oplus [u], \quad u \in \overline{K \cap Y_A}^{\sigma(X_A^{**}, X_A^*)}.$$

The zero-extension embedding $X_A^{**} \hookrightarrow X^{**}$ is weak-star to weak-star continuous, and the weak-star topology induced from X^{**} on the coordinate subspace X_A^{**} is exactly $\sigma(X_A^{**}, X_A^*)$. Therefore the same set K , regarded as a subset of X^{**} , is absolutely convex and weak-star compact. Moreover

$$K \subset Y \oplus [u], \quad u \in \overline{K \cap Y}^{\sigma(X^{**}, X^*)},$$

because $Y_A \subset Y$. This is precisely the (LK) witness required for X^* . Thus X^* is fully Mackey complete.

Verification and limitations

The result is a partial positive answer to Problem 4.15, not a full solution. It covers ℓ_p -sums, $1 < p < \infty$, of separable no- ℓ_1 spaces. The proof uses the countable-coordinate support structure of ℓ_p -sums in an essential way. It does not address arbitrary nonseparable, non-Asplund spaces without ℓ_1 .

The main point for review is Lemma 2. The Lindelof argument selects countably many elements of the original norming subspace at each countable coordinate stage; adding their supports ensures that, in the limit, those same elements lie in $Y \cap X_A^{**}$ and still norm the enlarged countable sub-sum up to a fixed constant.

The bounded novelty search for this run found no later literature answer to Problem 4.15 using the phrases “fully Mackey complete”, “fully Mazur”, and the exact wording of the problem. The local parsed corpus only showed the source paper and one later bibliography citation for the exact terminology.

Human-review recommendation

Treat this as a likely valid partial theorem. If the localization lemma survives review, this packet gives a new positive nonseparable class for Problem 4.15, including examples outside the separable and Asplund cases explicitly listed in the source paper.

References

- [1] A. J. Guirao, G. Martinez-Cervantes, and J. Rodriguez, *Completeness in the Mackey topology by norming subspaces*, J. Math. Anal. Appl. **478** (2019), no. 2, 776–789; arXiv:1812.10669.
- [2] H. P. Rosenthal, *A characterization of Banach spaces containing ℓ_1* , Proc. Nat. Acad. Sci. U.S.A. **71** (1974), 2411–2413.